Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · CHI-SQUARE INDEPENDENCE

are two categorical variables related?

Table 1. Contingency (observed [expected] (row% / col%) std. residual)

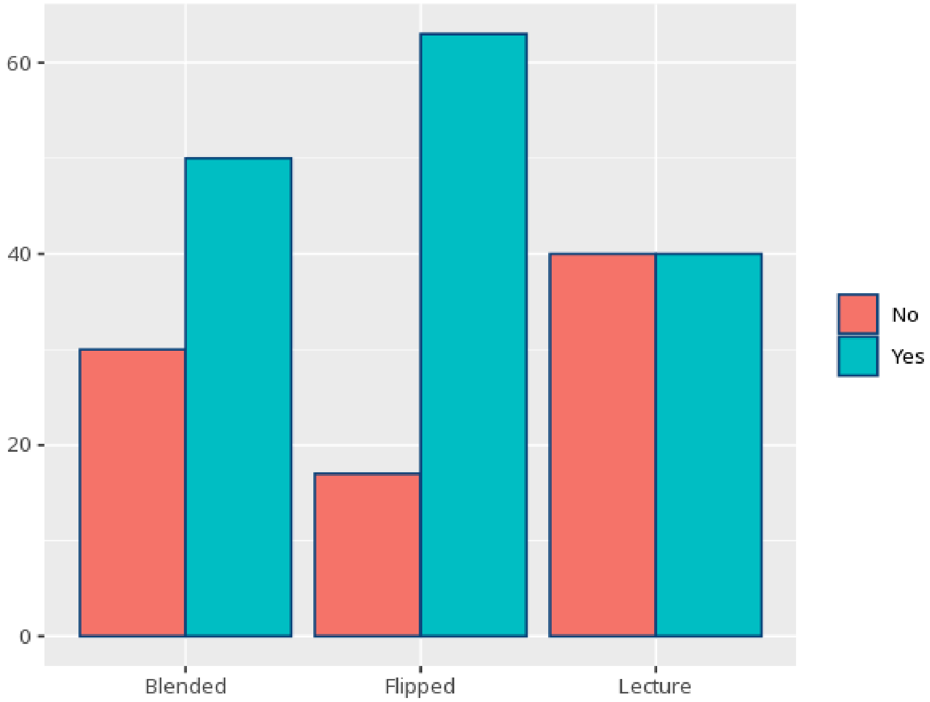
Row \ Column	No	Yes	Total
Blended	30 [29.00] (37.5% / 34.5%) r=0.28	50 [51.00] (62.5% / 32.7%) r=−0.28	80
Flipped	17 [29.00] (21.3% / 19.5%) r=−3.42	63 [51.00] (78.8% / 41.2%) r=3.42	80
Lecture	40 [29.00] (50.0% / 46.0%) r=3.13	40 [51.00] (50.0% / 26.1%) r=−3.13	80
Total	87	153	240

Table 2. Chi-square test

χ^2	df	p	Cramér's V [95% CI]
14.39	2	<.001	0.24 [0.13, 1.00]

the contingency table is r×c - columns expand to the number of categories in the column variable; each cell shows observed [expected] (row% / col%) and its standardized residual r (chisq.test()$stdres), with a warning (suggesting Fisher's exact) if expected counts are too small. A standardized residual beyond about ±1.96 flags a cell that significantly drives the association. For 2×2 tables chisq.test() applies Yates' continuity correction to the χ^2 by default (set correct=FALSE for the uncorrected χ^2); the standardized residuals are unaffected by the correction.

Figure. Cross-classification



Understanding the numbers

Row \ Column. The row-variable category heading each row of the contingency table. Here, the table has 3 row categories by 2 column categories.

Cell (observed [expected] / %s / residual). Each cell shows the observed count, the expected count under independence in brackets, its row % and column %, and its standardized residual r. A standardized residual (r) beyond about ±1.96 flags a cell that significantly drives the association; the smallest expected count here is 29.0.

Cell (observed [expected] / %s / residual). Each cell shows the observed count, the expected count under independence in brackets, its row % and column %, and its standardized residual r. As with the first column, every cell here is read the same way, across 2 column categories.

Additional columns. One more column per remaining category of the column variable, each read the same way. Here, the contingency table expands to 2 column categories in total.

Total. The row or column margin - the sum across that row or column. Here, the table's totals sum to N = 240.

χ^2. The size of the discrepancy between observed and expected counts if the two variables were truly independent. Here, $\chi^2(2, N=240) = 14.39$.

df. Degrees of freedom - (rows – 1) × (columns – 1) for this contingency table. Here, df = 2.

p. The probability of an association this strong (or stronger) if the two variables were truly independent. Here, $p < .001$ - below your significance threshold, the variables are related.

Cramér's V. The strength of association, on a 0 (none) to 1 (perfect) scale. Here, V = .24 [.13, 1.00].

How to read this test

Tests whether two categorical variables are associated. A p below alpha means they are related; Cramér's V gives the strength of association (0–1); each cell's standardized residual (r) beyond about ±1.96 flags a category combination that significantly drives the association. Valid only when expected counts are mostly ≥ 5 (the app flags this and suggests Fisher's exact) and each case appears once (raw counts, not percentages). Your significance threshold (α) is 0.05.

APA template: A chi-square test of independence gave $\chi^2(2, N=240)=14.39$, $p < .001$, V=.24 [.13, 1.00].

Statistical basis: Chi-square test of independence (Pearson, K. (1900). "On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling." Philosophical Magazine, 50(302), 157-175.); Cramér's V effect size (w-family; Cohen, 1988) (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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